

Effects of Walnuts on Endothelial Function in Overweight Adults with Visceral Obesity: A Randomized, Controlled, Crossover Trial



Objectives Metabolic syndrome is a precursor of diabetes and cardiovascular disease (CVD). Walnut ingestion has been shown to reduce CVD risk indices in diabetes. This randomized controlled crossover trial was performed to investigate the effects of daily walnut consumption on endothelial function and other biomarkers of cardiac risk in a population of overweight individuals with visceral adiposity. **Methods** Forty-six overweight adults (average age, 57.4 years; 28 women, 18 men) with elevated waist circumference and 1 or more additional signs of metabolic syndrome were randomly assigned to two 8-week sequences of walnut-enriched ad libitum diet and ad libitum diet without walnuts, which were separated by a 4-week washout period. The primary outcome measure was the change in flow-mediated vasodilation (FMD) of the brachial artery. Secondary measures included serum lipid panel, fasting glucose and insulin, Homeostasis Model Assessment–Insulin Resistance values, blood pressure, and anthropometric measures. **Results** FMD improved significantly from baseline when subjects consumed a walnut-enriched diet as compared with the control diet ($1.4\% \pm 2.4\%$ versus $0.3\% \pm 1.5\%$; $p = 0.019$). Beneficial trends in systolic blood pressure reduction were seen, and maintenance of the baseline anthropometric values was also observed. Other measures were unaltered. **Conclusion** Daily ingestion of 56 g of walnuts improves endothelial function in overweight adults with visceral adiposity. The addition of walnuts to the diet does not lead to weight gain. Further study of the potential role of walnut intake in diabetes and CVD prevention is warranted.